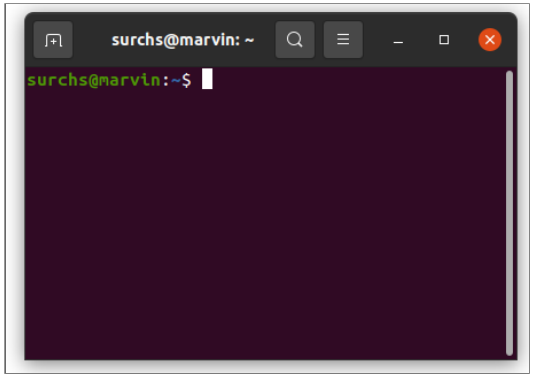
Introduction to the bash shell - exercises

- A brief lecture recap
- Exercises (p.22 onwards)

Note: Run bash commands in a terminal, not from the Jupyter notebook `.ipynb` file (for reference only).

Setting up for the exercises

1. Open your terminal



2. and confirm that you are running the **bash** shell:

In [ ]:

```
echo $0  # bash
echo $SHELL  # /bin/bash
```

3. **Copy** the data for the exercises (`shell-course`) out of the **QLS-course-materials** directory and into your **home directory** (`/home/USERNAME` or `/Users/USERNAME`):

In [ ]:

```
cp -r ~/QLS-course-materials/Lectures/2026/data/shell-course ~
```

A brief recap of the lecture

1. How to **navigate** files and directories
2. How to **modify and move** files and directories
3. How to **find** files and directories
4. Shell **variables** and scripts

Examples are provided in code snippets for you to try in your terminal as desired.

Bash command recap

In [ ]:

```
ls -l ~/shell-course
```

A shell command has 3 parts:

1. A **command** (`ls`),
2. One or more **options** (`-F`), also called a **flag** or a **switch**, and
3. An **argument** (`~/shell-course`)

It is generally recommended to follow the order: `command [options] <arguments>` (e.g., `head -n 1 myfile.txt`)
However, depending on your OS and versions of installed commands, the order of options and arguments might be less strictly enforced.
When in doubt, always specify options first, arguments second.

Recap: navigation

- The file system is responsible for managing information on the disk
- Directories can also store other (sub-)directories, which forms a directory tree
- `cd <path>` changes the current working directory
- `ls <path>` prints a listing of a specific file or directory; `ls` on its own lists the current working directory
- `pwd` prints the user's current working directory
- `/` on its own is the root directory of the whole file system
- A relative path specifies a location starting from the current location
- An absolute path specifies a location from the root of the file system
- `..` means "the directory above the current one"; `.` on its own means "the current directory"

Refresher `pwd`

`pwd` -> print working directory

In [ ]:

```
pwd
```

- `pwd` lets you know where you are
- It always prints absolute paths
- It's a great way to quickly get your bearings

Refresher `ls`

`ls` -> list directory contents

In [ ]:

```
ls -la
```

- `ls` lists the contents of the current working directory by default
- you can give it many options to change what is printed
- you can list other directories, by supplying them as arguments
- `.` stands for the current working directory
- `..` stands for the parent directory, the directory above the current directory
- file/directory names beginning with `.` are hidden from listing by default (here, the `-a` (`--all`) flag displays them)

Refresher `cd`

`cd` -> changes your working directory

In [ ]:

```
pwd
```

In [ ]:

```
cd ~/shell-course
```

In [ ]:

```
pwd
```

- `cd` changes the current working directory
- it expects a relative or absolute path to the new working directory you want to change to
- if you give it no argument, it will go back to your home directory

Refresher `home directory`

- Your home directory is where your user-specific files are
- on Linux it is in `/home/USERNAME`
- on Mac it is in `/Users/USERNAME`
- it contains your files and config files

In [ ]:

```
pwd
```

In [ ]:

```
cd
```

In [ ]:

```
pwd
```

In [ ]:

```
cd -
```

- `cd` without arguments brings you to your home directory
- `~` is a shorthand for your home directory -> `cd ~` also brings you there
- your home directory has a path -> `cd /home/USERNAME` also brings you there
- `-` is a shorthand for the directory you were in before the last `cd` call

Recap: modifying things

- `cp <old> <new>` copies a file (`cp -r` for a directory)
- `mkdir <path>` creates a new directory
- `mv <old> <new>` moves (renames) a file or directory
- `rm <path>` removes (deletes) a file (`rm -r` for a directory)
- `touch <file>` creates an empty text file or updates the access time of an existing file
- `*` matches zero or more characters in a filename, e.g. `*.txt` matches `a.txt` and `any.txt` (all files ending in `.txt`)
- `?` matches any single character in a filename, e.g. `?.txt` matches `a.txt` but not `any.txt`
- The shell does not have a trash bin: once something is deleted, it's really gone

Refresher `cp`

`cp` -> copy files and directories to a new path

In [ ]:

```
cd ~/shell-course/interesting_files
```

In [ ]:

```
ls
```

In [ ]:

```
cp the_meaning_of_life.txt the_meaning_of_life_backup.txt
```

In [ ]:

```
ls
```

- `cp` (and `mv` too) takes the form `cp [old-path] [new-path]`
- `cp` can operate on many files at once as long as the target is a directory
- The `-r` flag is needed to copy directories
- `cp` will keep the original file, whereas `mv` will move it
- `cp` and `mv` will overwrite without asking -> **dangerous**. The `-i` flag will make them ask first

Refresher `rm`

`rm` -> removes files and directories

In [ ]:

```
ls
```

In [ ]:

```
rm the_meaning_of_life_backup.txt
```

In [ ]:

```
ls
```

- `rm` generally deletes files without first asking
- `rm` deletes things **forever**. There is no trash-bin for bash and no undo button
- `rm` cannot delete directories without the `-r` flag

Recap: finding things

- we can print the structure of any given directory with `tree` (or `ls -R`)
- `find` is a great tool to search for **files and directories** based on their name and other meta-data like size, age, and so on
- `grep` is a great tool to search **within (text)files** for occurrences of a given string or even complex regular expressions
- pipes (`|`) allow us to combine the output of one command with the input of another command

Refresher `find`

`find` -> find files and directories by name and meta-data

In [ ]:

```
cd ~/shell-course
find . -name "my*"
```

- `find` is great to find all files with a certain name pattern
- `find` can also search for attributes like size and age

Refresher `grep`

`grep` -> find a text pattern inside of files and print the matches

In [ ]:

```
grep "rabbit" flying_circus/* --max-count 2
```

- `grep` is great to search for something **inside** of files
- `grep` can search for a simple string or complex regular expressions
- `grep` can be useful to extract lines with specific content out of a file

Refresher pipes

The `|` character is a pipe. It can be used to link the output of one bash command to the input of another bash command. Commands linked in this way are called pipelines

In [ ]:

```
grep "rabbit" flying_circus/*.txt --max-count 2 | wc --chars
```

- shell commands generally do one thing well
- linking commands can achieve powerful pipelines
- here `grep` finds text files and pipes the output to `wc` a program to count the number of characters and lines in a text
- we then get the total number of characters found by `grep`
- `>` and `>>` are special characters that can redirect output into files (we'll see this in a moment)

Refresher help

The bash shell has many great helper tools. Often they can answer questions without the need for Google:

- `--help` -> a flag that provides the basic usage and options for many bash commands
- `whatis` -> provides a brief description of a command
- `man` -> opens the manual for a given command, with comprehensive documentation of functionality, options, and usage examples
- `which` -> tells you where the program is located that is called by a command

In [ ]:

```
whatis wc
```

In [ ]:

```
wc --help
```

In [ ]:

```
which wc
```

Recap: variables and scripts

- the `$PATH` variable defines the directories where the shell will look for commands
- you can change `$PATH` for just your current shell session, or for all sessions (in `~/.bashrc` - do this with caution!)
- the `$` character is necessary to refer to the **value** of a bash variable
- often it makes sense to put the variable name inside curly braces `${ }` when referencing the value, to differentiate it from other text
- variables must be exported to be made accessible inside other scripts/programs: `export VARIABLE_NAME`
- shell scripts are executable text files that contain shell commands
- scripts need execution permission that we can give with the `chmod` command
- scripts start with the "shebang": `#!/bin/bash` that specifies which shell the script should be interpreted by

Recap of topics

1. How to **navigate** files and directories (`ls`, `cd`, `pwd`)
2. How to **modify and move** files and directories (`cp`, `mv`, `rm`)
3. How to **find** files, directories and help (`find`, `grep`, `tree` and `man`, `whatis`, `which`)
4. Shell **variables** and scripts (`$PATH`, `.bashrc` , `${MY_VAR}`, `export`)

Exercise 1 - moving things around

Let's navigate to the `dir_of_doom` with `cd` and take a look inside with `ls` and `tree`.

In [ ]:

```
cd ~/shell-course/dir_of_doom
```

In [ ]:

```
tree
```

In [ ]:

```
ls -Rla
```

All these files are in `the_wrong_dir`, let's move them somewhere else.

Exercise 1a

Move all the files from `the_wrong_dir` to a new directory called `the_right_dir`, without moving each file individually. Recall the form of the `mv` and `cp` commands: `mv [old_path] [new_path]`.

Hints

- the `mv` command can move many files at once, as long as the `[new_path]` is a directory and not a file
- e.g., `mv file1.txt file2.txt target_directory` works, but `mv file1.txt file2.txt file3.txt` does not
- a wildcard expands to match multiple file names. It has the same function as typing all the file names by hand
- `*` (asterisk) will match any character 0 or more times
- `?` (question mark) will match any character exactly once

Try it out!

In [ ]:

```
mkdir the_right_dir
mv the_wrong_dir/my_file?.txt the_right_dir
```

Exercise 1b

Now that everything in `the_right_dir` looks good, remove the `the_wrong_dir`.
Try it out!

Recall:

- `rm` can remove files
- `rm` can only remove directories with the `-r` ("recursive") flag

In [ ]:

```
rm -r the_wrong_dir
```

Note: there is no way to get `the_wrong_dir` back! Be **very careful** with `rm` commands, especially when using wildcards and relative paths

Summary

- `cp old new` copies a file
- `mkdir path` creates a new directory
- `mv old new` moves (renames) a file or directory
- `rm path` removes (deletes) a file
- `*` matches zero or more characters in a filename, so `*.txt` matches all files ending in `.txt`
- `?` matches any single character in a filename, so `?.txt` matches `a.txt` but not `any.txt`
- The shell does not have a trash bin: once something is deleted, it's really gone

Exercise 2 - pipes

Now let's take a look in the `flying_circus` directory. There we have several text files and we want to know what the shortest text file is. Here we can make use of several tools:

- `wc`
- `sort`
- `head`

You can use the `whatis` command to check what they do.

Exercise 2a

Navigate to the `flying_circus` directory, and print the **number of lines** of each text file in the directory using a single `wc` command.

Hints

- A wildcard (`*`) expands to match multiple file names
- The `--help` flag shows all options you can use to control the behaviour of a command

Try it out!

In [ ]:

```
wc -l *.txt
```

Exercise 2b

Instead of printing the `wc` command output to the screen (called STDOUT), **redirect** the output into a file `file_length.txt`.

Hints

Some special characters that redirect the output normally printed to STDOUT:

- `|` ("pipe") redirects the output to a second bash command as input
- `>` redirects the output to a file and **overwrites** whatever is in the file
- `>>` redirects the output to a file and **appends** to this file if it exists

Try it out!

In [ ]:

```
wc -l *.txt > file_length.txt
```

In [ ]:

```
cat file_length.txt
```

Now let's **sort** the text in `file_length.txt` by the number of lines:

In [ ]:

```
sort file_length.txt
```

Note: If the file lengths were not sorted correctly, it is because `sort` interprets numbers as text by default. You can also see this by running `sort` on the contents of `dangerous_rabbits.txt`.
`sort --help` tells us that `--numeric-sort` explicitly ensures numerical value sorting.
Redirect the numerically-sorted output into a file called `sorted_length.txt`.

In [ ]:

```
sort --numeric-sort file_length.txt > sorted_length.txt
```

Now let's read only the first line of `sorted_length.txt`, to find the name of the shortest text file in the `flying_circus` directory.

In [ ]:

```
head -n 1 sorted_length.txt
```

Exercise 2c

So far, to find the shortest text file we have run:

1. `wc -l *.txt > file_length.txt`
2. `sort --numeric-sort file_length.txt > sorted_length.txt`
3. `head -n 1 sorted_length.txt`

This created 2 extra text files and took 3 commands. This is a good use for bash pipelines!

Recall: the `|` (pipe) character redirects the output to another command as input.

Delete the intermediate files you created (ending in `_length.txt`). Then, rewrite the 3 commands into one pipeline command using `|`, so that the name of the shortest file is printed without creating any files.

Try it out!

In [ ]:

```
rm *_length.txt
wc -l *.txt | sort --numeric-sort | head -n 1
```

Summary

- `|` the "pipe" character redirects the output to a second bash command as input. e.g. `wc -l *.txt | head -n 1`
- `>` redirects the output to a file and **overwrites** whatever is in the file. e.g. `wc -l *.txt > file_lengths.txt`
- `>>` redirects the output to a file and **appends** to this file if it exists. e.g. `wc -l *.txt >> file_lengths.txt`

Exercise 3 - grep

Some of the text files in the `flying_circus` directory are so long because they contain the complete scripts to Monty Python movies. `brian.txt` contains the script to "The Life of Brian". Let's make a personalized script for the actor playing the role of `"Brian"` - containing only the lines said by that role.
For this we can use `grep`, which can search for text snippets (i.e. strings) **inside** of files.
Let's first look for `"Brian"`:

In [ ]:

```
ls
```

In [ ]:

```
grep "Brian" brian.txt
```

Now we get every line containing `"Brian"`. But we only want the lines that the character Brian *says*.
For this, we can:

- search specifically for the string `"Brian:"` including the `:` character
- use the `^` (caret) character to only find occurrences that are **at the beginning of the line**, i.e. `^Brian:`
    - FYI: the `man` pages for `grep` have a section dedicated to these patterns, called regular expressions

Let's add these to our `grep` command and then redirect the output to a new file called `my_lines/Brians_lines.txt`. Then, check that it worked correctly.

In [ ]:

```
mkdir my_lines
```

In [ ]:

```
grep "^Brian:" brian.txt > my_lines/Brians_lines.txt
```

In [ ]:

```
head my_lines/Brians_lines.txt
```

Now, what if we wanted to:

- record the exact command we used to create the lines for this role?
- re-run the exact same command in the future, e.g. to re-create the lines for the "Brian" role?
- create the lines for another role in this movie?
- be able to change the role we create lines for without modifying commands?

Exercise 3 - scripts

This is a good use case for shell scripts! We can:

1. put the commands we have just written in a shell script to re-run them again later
2. use a variable to store the name of the role so we can easily change what actor we generate lines for

Let's look at these steps separately.

Recap scripts

A **script** is a text file that contains shell commands and:

1. commonly has the `.sh` file ending to show it is a script
2. starts with the shebang: `#!/bin/bash` that specifies the shell that should run the script
3. has execution permission. This can be given with the `chmod +x` command

Let's check an example script in the `interesting_files` directory:

In [ ]:

```
ls -lF ../interesting_files/
```

In [ ]:

```
../interesting_files/run_me.sh
```

Exercise 3a

Create a script that runs our `grep` command to create the lines for the role of "Brian"

Steps:

1. In the `~/shell-directory/flying_circus` directory, create an (empty) script file called `create_lines.sh`
2. Copy the `grep` command to the shell script using any text editor (or, try using `echo` and `>` to redirect the entire original command into the file)
3. Add all the necessary shell script elements

In [ ]:

```
grep "^Brian:" brian.txt > my_lines/Brians_lines.txt
```

Try it out!.

HINTS:

- `touch <filename>` creates an empty file
- if you copy by hand, use the context menu to paste (right-click). `CTRL+C` is reserved in `bash` to cancel commands
- in `nano`, remember to save (write out) the file before you exit.
    - `^` for CTRL: `^G` means "press and hold CTRL together with the `G` key"
    - `M` for ALT: `M-U` means "press and hold ALT together with the `U` key"
    - Write out then is CTRL+`O`

In [ ]:

```
nano create_lines.sh

# OR
echo \
  'grep "^Brian:" brian.txt > my_lines/Brians_lines.txt' > create_lines.sh
# and then nano create_lines.sh to add shell script elements
```

Now let's

- give the script execution permission with `chmod +x`
- see if this worked with `ls -lF` (executable files are marked with `*`)
- remove the existing `my_lines/Brians_lines.txt` file and run our script with `./create_lines.sh`

In [ ]:

```
ls -lF
```

In [ ]:

```
chmod +x create_lines.sh
```

In [ ]:

```
ls -lF
```

In [ ]:

```
rm my_lines/Brians_lines.txt
./create_lines.sh
```

Exercise 3 - variables

We can use variables to change the role that the script creates the lines for.

Recap variables

- To *define* (assign a value to) a variable we use the `=` character
- To *access* the value of a variable we use the `$` character
- A newly defined variable is a **shell variable** that is not visible to other programs we start from our shell
- With the `export` command, we can turn our variable into an **environment variable** that is visible to other programs

Exercise 3b

1. Create a shell variable called `ROLE` and assign the name of a different role, `"Vendor"`, as the value
2. Confirm the value was correctly assigned with `echo`
3. Turn our variable `ROLE` into an **environment variable** so our script can see it
4. Confirm that `ROLE` is now part of the environment variable list using `printenv` and `grep`

Try it out!.

In [ ]:

```
ROLE="Vendor"
```

In [ ]:

```
echo ${ROLE}
```

In [ ]:

```
export ROLE
```

In [ ]:

```
printenv | grep ROLE
```

Exercise 3c

Finally, replace the hard-coded value `"Brian"` in the script `create_lines.sh` with the new variable `ROLE`.
To confirm it worked, try changing the value of the `ROLE` variable in the shell and then re-run `create_lines.sh`. Some other roles to try:

- Baby
- Balthasar
- Eremite
- Door

Hints

- `$` to access the value of the variable
- variable names are case sensitive
- wrapping the variable in `{ }` is recommended when the variable is surrounded by other text, so bash knows where the variable name ends
    - we can embed a variable in a string like this: `"Hello World!"` -> `"Hello ${PLACE}!"`

Try it out!

In [ ]:

```
cat create_lines.sh
```

In [ ]:

```
nano create_lines.sh
# OLD: grep "^Brian:" brian.txt > my_lines/brians_lines.txt
# NEW: grep "^${ROLE}:" brian.txt > my_lines/${ROLE}s_lines.txt
```

In [ ]:

```
ROLE=Door
```

In [ ]:

```
./create_lines.sh
```

Summary

- `grep` is useful for searching within (text)files for occurrences of a given string or even complex regular expressions
- Shell scripts are a very powerful way to automate, repeat and document steps
- Variables can store values that scripts operate on
- We access the value of variables with `$` and we can turn variables into environment variables with `export`
- Here, we knew the variables used inside the script and changed those directly in the environment. In practice, there are easier ways to tell our script the actor we want to create lines for (e.g., we could make the script accept arguments like other bash commands)

Exercise 4 - a neuroimaging dataset example

When working with research datasets, commands like `head`, `tail`, `cat`, `wc`, and `grep` are extremely useful to quickly inspect large tabular data files (e.g., `.csv` and `.tsv`) without having to load them in Excel, Google Sheets, etc.
The file `participants_nbsub-100.tsv` inside the `shell-course` directory is a tab-separated table containing data for 100 subjects from the ABIDE dataset. Each row represents a subject except for the first row, which contains the column names.

Exercise 4

Explore `participants_nbsub-100.tsv`, **without opening the file**:

1. Print only the first row (the column names) to see what kinds of subject data are available
2. Confirm that besides the header, there are 100 rows (lines of subject data) in the file
3. Print just the information for the subject `50012`
4. There's a column `SITE_ID` which has 3 possible values: `PITT`, `OLIN`, `OHSU`. Find just the rows that have the value `"PITT"`, and output those rows to a new file `participants_PITTonly.tsv`

Hints

- using `-n +NUM` (note the `+`) with `tail` returns all lines in the file **starting from** line `NUM`
- `wc -l` counts the number of lines in a file

Try it out!

In [ ]:

```
head -n 1 participants_nbsub-100.tsv
```

In [ ]:

```
tail -n +2 participants_nbsub-100.tsv | wc -l
```

In [ ]:

```
grep 50012 participants_nbsub-100.tsv
```

In [ ]:

```
grep PITT participants_nbsub-100.tsv > participants_PITTonly.tsv
```

Exercise 5 - the `$PATH` variable

When we run scripts that we have created (like `create_lines.sh`), we need to specify the path to the script: `./create_lines.sh` (recall that `.` means "current directory").
We've learned that when you type a command, the shell will go and search for executable files with this name in a number of directories, defined in the `$PATH` variable:

In [ ]:

```
echo $PATH
```

Unless a script is in one of these directories, the shell won't find it.
We have some scripts inside the `interesting_files` directory, but this directory is not in `$PATH`. We cannot run them without specifying their path:

In [ ]:

```
ls -lF interesting_files
```

In [ ]:

```
run_me.sh
```

Exercise 5a

How can we add the `interesting_files` directory to the `PATH` variable?
Just like the `ROLE` variable in the previous exercise, we can **re-assign** the value of the `PATH` variable.

Hints:

- In `$PATH`, directories are separated by a `:` character
- *Append* another directory to the list currently in `$PATH` using the `:` delimiter, and then re-assign the result to the `PATH` variable

Try it out!

In [ ]:

```
echo $PATH
```

In [ ]:

```
PATH=${PATH}:~/shell-course/interesting_files
```

In [ ]:

```
echo $PATH
```

Let's see if the shell can now find our script:

In [ ]:

```
run_me.sh
```

Modifying `$PATH` in the shell only affects the current shell session! To make this permanent, we would need to make the change in `~/.bashrc`.

Exercise 5b

Create a new directory `~/shell-course/my_scripts`. Inside, create a script called `favorite_color.sh` that, when run, prints the text:

```
My favorite color is: <FAVORITE COLOR>
```

where `<FAVORITE COLOR>` can be set by an **environment variable**.
Then, make the necessary changes to `$PATH` so that the script can be run just by calling the filename: `favorite_color.sh`

Try it out!

Hints

- `echo` prints text to the screen
- Recall the key components of a shell script: `#!/bin/bash`, execute permissions, and commands to run
- You can use single quotes `''` to prevent the Bash from interpreting reserved characters (e.g., `#`, `$`) in a string
- `export` turns a shell variable into an environment variable
- You can edit the `$PATH` variable by reassigning it

In [ ]:

```
mkdir ~/shell-course/my_scripts
cd ~/shell-course/my_scripts
```

In [ ]:

```
echo '#!/bin/bash' > favorite_color.sh
echo "echo My favorite color is: ${FAVORITE_COLOR}" >> favorite_color.sh
```

In [ ]:

```
chmod +x favorite_color.sh
```

In [ ]:

```
PATH=${PATH}:~/shell-course/my_scripts
```

In [ ]:

```
export FAVORITE_COLOR=yellow
```

In [ ]:

```
cd ~
favorite_color.sh
```

Summary

- there are two types of variables: "shell variables" and "environment variables"
    - only environment variables get passed to programs you call from the shell
    - you can turn a shell variable into an environment variable with `export`
- to retrieve the value of a variable, we need the `$` character (e.g. `$VAR` vs `VAR`)
- the shell will look for programs in your command in directories defined in the `$PATH` variable
- `$PATH` and other environment variables are set by startup files at the system and user level
- you can edit the startup files for your user in your home directory (e.g. `~/.bashrc`)

Final tips

The Bash shell is very powerful for:

- automating many repetitive tasks, particularly when they involve manipulating files and directories
- documenting executed commands and programs (through scripts) so they can be re-used
- coordinating different (in)dependent processes together (pipelines)
- accessing remote computers like Digital Research Alliance of Canada (Compute Canada) clusters
- accessing and understanding many tools in the neuroimaging world

However, `bash` is not the best tool for things like:

- complex operations on structured data (tables, dictionaries, etc.)
- performing complex statistical tasks like the ones needed in research data analyses
- data visualization

For these tasks, modern programming languages like `python` offer better and more user-friendly tooling, control flow, debugging, and other features.

What else can I do?

Check out the documentation for some other useful commands:

- **rsync**: local and remote file transfer (synchronization) that can detect and transfer differences only
- **tmux** (see also **this beginner's guide**): manage multiple terminal "windows" and keep sessions running in the background
- `cat -e` and **dos2unix**: check and convert line endings between Windows/Mac/Linux file formats (very useful if you work on a remote server that is a different OS)
- `sed` and `awk` **tutorial**: more advanced string manipulation/replacement and selecting specific sections of text
    - often used in a context with `grep`

References

There are lots of excellent resources online for learning more about bash:

- The GNU Manual is *the* reference for all bash commands:
  http://www.gnu.org/manual/manual.html
- "Learning the Bash Shell" book:
  http://shop.oreilly.com/product/9780596009656.do
- An interactive on-line bash shell course: **https://www.learnshell.org/**
- The reference page of the software carpentry course:
  https://swcarpentry.github.io/shell-novice/reference.html